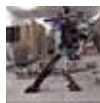




## Word Flow on iOS

Microsoft is bringing its excellent Windows phone keyboard to iOS  
<http://dgit.in/iWinKeybd>



## Here's Atlas

Forget iRobot Roomba and Samsung's POWERbot, this is what a robot cleaning your house might look like. Or maybe not.  
<http://dgit.in/RoboAtlas>

From the labs

played back in radio towers and anyone with a radio scanner in the vicinity of the phone signal was able to eavesdrop.

### Fact

*Bell Labs developed the AMPS or 1G standard and Motorola came out with the device to support it. This became the first commercially available mobile phone, the DynaTAC 8000X.*



## 2G

To overcome the problems of 1G, a new standard 2G or GSM (Global Systems for Mobile communications) was introduced in 1991. Offering digitally encrypted phone calls, this standard was more secure and efficient than its predecessors, hence it was adopted world-wide. A few years later in 1995, SMS services were integrated within the technology which made it an even better standard. With 2G, phones became much smaller, battery efficiency was improved and voice clarity was better. But the times were changing and the internet was at the cusp of booming. Sadly, 2G with its 64Kbps theoretical speed of data transfer was not up for the task. This was the main hurdle in the research and development of the third generation.

### Fact

*The original iPhone featured cellular connectivity using GPRS and EDGE bands.*

## 2.5G and 2.75G

Between the second and third generations, lies GPRS (General Packet Radio Service) and EDGE (Enhanced Data rates for GSM Evolution) which are otherwise known as 2.5G and the 2.75G standards respectively. These transitional standards were later developed over the older 2G standard and basically provided more data speeds than 2G which was good enough to check email or load a webpage, but still fell considerably short of the network speeds 3G could achieve.

## 3G

In 2001, the age of 3G dawned. It was a big upgrade from the 2G standard but it was an upgrade. Nonetheless, the new standard had upgraded audio codecs and some part of the older GSM architecture. 3G also brought a new standard called the IMT-2000 (IMT – International Mobile Telecommunication). With the introduction of 3G, things got complex because there are many technologies which are compliant with the IMT-2000 standard, but the one most important is UMTS. UMTS, otherwise known as the Universal Mobile Telephone System, is the widely used 3G standard offering global roaming as it is globally recognised by a majority of countries. The upgrade in audio codecs saw enhanced audio and video



streaming, higher browsing speeds, faster data speeds, GPS support, video conferencing support and support for IPTV (TV through internet).

## 3.5 and 3.75G

Things started to evolve rapidly afterwards as the demand for faster internet speeds emerged. Soon, the 3.5G and 3.75G standards came into existence. These two standards otherwise known as HSPA and HSPA+ offered peak theoretical downlink speeds of 14Mbps and 168Mbps respectively. This was more than the 2Mbps downlink speeds 3G used to offer. While HSPA was metaphorically 3G on steroids, 3.75G was able to achieve much higher speeds and lower latency due to a new technology called MIMO (Multiple-input and Multiple-output), which set the ground for the 4G and LTE standards.

### Fact

*Some mobile operators are selling the 3.75G or HSPA+ plans as 4G because the real life speeds of the two are very close.*

## 4G

4G, otherwise known as IMT Advanced, is a complete overhaul from the earlier standards. 4G is currently deployed in two forms, one is WiMAX (Worldwide Interoperability for Microwave Access) and LTE (Long Term Evolution). Out of the two, LTE, has a better adoption rate across the globe. It is also due to the fact that WiMAX supports voice calling via VoIP only and requires new deployments, whereas LTE is made to work with existing deployments. LTE is also slightly faster than WiMAX.

**Key point:** LTE is the long term evolution of UMTS which was upgraded over the 2G standard.

LTE-Advanced is the latest chapter in 4G. It's faster than 4G and offers theo-



retical peak download speeds of upto 450Mbps over the theoretical peak download speeds of upto 300 Mbps achievable by LTE. This was a good enough reason to be excited about 4G finally coming to India. Although, you won't get the theoretical speeds due to various physical and technical factors, it is still good enough to be excited.

## 5G = 2G + 3G + 4G + Wi-Fi + ?

By now you must have realised one thing. Every new generation is better than the previous one, and not by a small margin. With 5G, the theoretical download speeds are expected to hit 10Gbps, making it 100 times faster than 4G. Also, like all previous generations, 5G will incorporate all its previous generations within itself. Additionally, it will also include specific new technology architectural changes that haven't been defined yet. Wi-Fi technology is also part of this heterogeneous equation and will play an important part in it. Wi-Fi derived technology will help the